

AMTI (NMTTC) - 2011

KAPREKAR CONTEST - FINAL - SUB-JUNIOR LEVEL

- Find all three digit and four digit natural numbers such that the product of the digits is a prime number. Find the sum of all such three digit numbers and the sum of all such four digit numbers. Find the biggest prime factor of each sum.
- Let a sequence of numbers be denoted as $t_1, t_2, t_3, \dots$, where $t_1 = 1$ and $t_n = t_{n-1} + n$. (n is a natural number). Find $t_2, t_3, t_4, t_{10}, t_{2011}$.
- When written out completely 16^{2011} has m digits and 625^{2011} has n digits. Find the value of $(m + n)$.
- Four digit numbers are formed by four different digits a, b, c, d (none of them is zero) without any repetition of digits. Prove that when the sum of all such numbers when divided by the sum of the digits a, b, c, d , the quotient is 6666.
- ABCD is a parallelogram. Through C a straight line is drawn outside the parallelogram. AP, BQ, DR are drawn perpendicular to this line from A, B and D. Prove that $AP = BQ + DR$.
- Falguni puts 12 plastic bags inside another plastic bag. Each of these 12 bags is either empty or contains 12 othe plastic bags. All together if 12 bags were non-empty, find the total number of bags.
- A, B and C are the digits of a three digit number ABC ($A, C \neq 0$). The number got by reversing the digits (also a three digit number) is added to ABC and the sum is found to be a square number. Find all such three digit numbers.
- Nine square are arranged to form a rectangle ABCD. The smallest square P has an area 4 sq. units. Find the areas of Q and R.


